

Response to Referee 2

We thank Referee 2 for the thoughtful and constructive report. The comments have helped us clarify key modeling assumptions, better position the paper within the relevant strands of the literature, and strengthen the discussion of external validity and fiscal reaction functions. Below we respond point by point. For ease of reference, we quote the referee’s comments in italics and then provide our response.

1. Exogenous commodity production

Referee comment

Treating commodity production as fully exogenous limits the model’s realism, especially for economies where output is shaped by investment cycles, geological constraints, and infrastructure bottlenecks. This assumption should be acknowledged explicitly as a caveat, and its implications discussed in more depth.

Response

We thank the referee for highlighting this important modeling assumption. Oil production is treated as an exogenous stochastic process. This choice is motivated by two empirical regularities for Ecuador over the estimation period (2004–2019): (i) quarterly oil output is highly persistent and evolves at relatively low frequency compared to other macro aggregates; and (ii) short-run changes in oil production are only weakly correlated with domestic non-oil economic activity, whereas oil prices and global conditions are more salient drivers. These features indicate that short-run oil output fluctuations are limited and persistent, and therefore can be parsimoniously represented as an exogenous process in the model, and they align with the treatment of commodity production in several small open economy DSGE models for resource exporters (e.g., Medina and Soto, 2016; Melina, Yang and Zanna, 2016; Berg et al., 2012).

We fully agree, however, that this abstraction should be clearly flagged and its implications discussed. In the revised manuscript, we therefore make the following changes:

- In the model description section, we explicitly note that commodity production is modeled as exogenous and explain that this abstraction is chosen to keep the focus on the transmission of commodity *revenue* shocks through the fiscal block, external debt, and the country risk premium, rather than on the determinants of oil-sector investment and depletion. See p. 10, 3rd paragraph, and footnote 12.
- After the description of the fiscal simulation results, we add a footnote discussing how endogenizing oil production could affect our results (see Section 6, footnote 33, p. 27). In particular, we note that allowing for capital accumulation in the oil sector and investment decisions responding to expected prices would likely amplify the long-run volatility of government oil revenues. Under adverse oil-price paths, endogenous disinvestment could lead to sharper and more persistent revenue shortfalls, further worsening fiscal balances and raising the risk premium. Conversely, under favorable oil-price paths, endogenous investment could strengthen the improvement in the public finances. In both cases, the underlying asymmetry that we document—namely that identical fiscal actions can lead to very different consolidation outcomes depending on the realization of commodity revenues—would remain, and could plausibly be magnified.

We emphasize that extending the framework to model the oil sector explicitly is feasible but falls outside the scope of the current paper, whose main objective is to quantify the macro-fiscal consequences of commodity revenue uncertainty for a given consolidation program. We now state this more clearly in the revised text and present the exogenous-production assumption as a deliberate simplification rather than an innocuous modeling detail.

2. Use of unexpected fiscal shocks vs anticipated consolidations

Referee comment

The simulations rely on sequences of unanticipated shocks to replicate the programmed fiscal paths. This modelling choice understates anticipatory behaviour, which is important in practice because consolidation measures are typically debated, negotiated, and signaled well in advance. A discussion of this limitation in relation to the existing DSGE literature on anticipated versus unanticipated fiscal adjustments would strengthen the paper.

Response

We thank the referee for this important point. We agree that modeling a fiscal consolidation as a sequence of unanticipated shocks understates anticipatory behavior, which is empirically and theoretically relevant given that consolidation measures are typically announced, debated, and phased in over time. Anticipation can affect private consumption, investment, and labor supply well before policy measures are implemented, and ignoring this channel may distort the dynamic responses.

In the original version of the paper, the consolidation paths were implemented using sequences of unexpected shocks in the linearized model. This approach allowed us to condition on the fiscal trajectories specified by the program but did not permit forward-looking agents to internalize future policy changes. As the referee correctly notes, this abstracts from anticipation effects emphasized in the DSGE literature on fiscal policy.

In the revised paper, we address this limitation directly by switching to a nonlinear perfect-foresight framework. The IMF-supported consolidation program is now implemented as a fully anticipated sequence of deterministic fiscal instruments. Agents observe the entire quarter-by-quarter paths of tax rates, government expenditure, and oil revenue over the consolidation horizon, and the model is solved for the resulting transition dynamics. This approach preserves nonlinearities and explicitly incorporates anticipatory responses, in line with the treatment of announced fiscal adjustments in the DSGE literature.

We clarify this change in the main text and document the implementation in detail (see Section 6, p. 21, 1st paragraph). As a result, the revised simulations capture both the contemporaneous and anticipatory effects of consolidation measures, while maintaining a clear link between the estimated structural parameters and the policy experiments. The main qualitative results continue to hold, but the new framework provides a more realistic representation of the timing and transmission of fiscal consolidations.

3. Fiscal rule specification

Referee comment (excerpt)

The estimated fiscal rules governing government consumption, investment, and transfers play a central role in shaping the dynamics, yet receive limited discussion. The authors should provide greater justification for the chosen functional forms and consider discussing robustness. This could be linked more explicitly to DSGE implementations of fiscal rules in commodity exporters (e.g., Medina and Soto, 2016) and to empirical work on structural-balance frameworks (e.g., Frankel, 2011).

Response

We thank the referee for this helpful suggestion. We agree that the fiscal reaction functions for government consumption, public investment, and transfers deserve a clearer discussion, particularly because they matter for the model’s stochastic dynamics and Bayesian estimation. In the revised manuscript we therefore expand the model description to motivate the functional form and relate it to standard DSGE implementations used for emerging and commodity-exporting economies (e.g., spending rules that respond to output and to deviations of debt from a target, as in Medina and Soto, 2016; Melina, Yang and Zanna, 2016; Berg et al., 2012). We also link the interpretation of the estimated coefficients to the empirical evidence on procyclicality in resource-rich economies and to work on structural-balance frameworks (e.g., Frankel, 2011). See p. 11, last paragraph.

Importantly, we now clarify in the text that these estimated reaction functions are used to discipline policy behavior in the estimation sample, but they do not determine the fiscal consolidation experiment. In the consolidation simulations, fiscal instruments (including spending components and transfers) are imposed as deterministic, fully anticipated paths consistent with the IMF agreement (now implemented under perfect foresight, as discussed in the response to your previous comment), so the consolidation results are driven by the program’s instrument trajectories interacting with debt dynamics and the risk-premium channel, rather than by the estimated fiscal rules.

We add a brief paragraph explaining this separation between (i) the estimated rules used for identification and (ii) the deterministic program implementation used for policy counterfactuals (see p. 21, 1st paragraph, and footnote 26). This addresses the referee’s concern about the role of fiscal rules while keeping the focus on the paper’s main objective: quantifying the sensitivity of consolidation outcomes to commodity-revenue realizations.

4. External validity

Referee comment

Although the authors argue that the framework applies broadly to commodity exporters, the model is tightly calibrated to Ecuador’s institutional and macroeconomic structure. A brief reflection on how the results might differ in countries with alternative tax systems, exchange-rate regimes, fiscal anchors, or debt-management frameworks—such as those modelled in LIC/EM DSGE studies (e.g., Bi, Shen and Yang 2016)—would strengthen the external relevance of the findings.

Response

We thank the referee for this suggestion and agree that a more explicit discussion of external validity will help readers interpret our findings. In the revised manuscript, we add a paragraph in the conclusion explaining how the strength of the main mechanism may vary across commodity exporters with different institutions (see p. 29; last paragraph). We emphasize that while our quantitative magnitudes are Ecuador-specific, the qualitative channel—commodity-revenue realizations

affecting debt and sovereign spreads, thereby shaping macro outcomes under a given consolidation plan—applies broadly. We discuss how this channel may be attenuated under monetary autonomy and flexible exchange rates, mitigated by stronger fiscal anchors or stabilization mechanisms (e.g., structural-balance frameworks), and amplified in environments with tighter fiscal limits or steeper debt–spread relationships (as in related LIC/EM DSGE studies).

5. Engagement with the small open economy literature

Referee comment

The paper would benefit from a more explicit link to the broader small open economy literature in commodity-rich settings. Existing empirical and modelling work—such as Roch (2017) for Chile, Colombia, and Peru; Knop and Vespignani (2014) and Vespignani (2015) for Australia; and Al Jabri et al. (2022) for Oman—provides useful benchmarks. Situating the analysis within this literature would clarify how the paper extends or complements established findings on commodity-driven macroeconomic and fiscal dynamics.

Response

We agree that strengthening the link to the broader small open economy literature on commodity-rich settings would improve the paper. In the revised Related Literature section, we now situate our contribution more explicitly relative to empirical evidence on commodity shocks and fiscal dynamics in both emerging and advanced commodity exporters; see p. 3, last paragraph.

First, we discuss work documenting the macroeconomic adjustment to commodity terms-of-trade shocks in Latin American small open economies, such as Roch (2017) for Chile, Colombia, and Peru, which highlights the roles of exchange-rate flexibility and fiscal space. Second, we reference evidence for advanced commodity producers, including sectoral analyses for Australia (Knop and Vespignani, 2014) documenting heterogeneous industry responses to commodity price shocks, and related work showing heterogeneity in macroeconomic transmission across regions (Vespignani, 2015). Third, we incorporate empirical evidence for oil exporters such as Oman (Al Jabri, Raghavan and Vespignani, 2022), which documents the strong influence of oil prices on fiscal aggregates and emphasizes fiscal smoothing using buffers and debt.

We clarify that our paper complements this empirical literature by offering a structural DSGE framework with a detailed fiscal block and a risk-premium channel, and by evaluating a concrete IMF-supported consolidation program under realistic commodity-revenue uncertainty. This helps make clear how our analysis extends established findings on commodity-driven macroeconomic and fiscal dynamics.

Concluding remark

We are grateful to Referee 2 for the insightful comments and suggestions. The changes described above—clarifying key modeling assumptions, introducing perfect-foresight simulations that capture anticipation effects, discussing fiscal reaction functions in more depth, expanding the treatment of external validity, and strengthening the links to the broader small open economy literature—have, in our view, substantially improved the clarity, credibility, and relevance of the paper.

Sincerely,
Manuel González-Astudillo, Juan Guerra-Salas, Avi Lipton

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